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Revolutionizing Reservoir Access: The World's First Capstan Conveyance with Polymer-Locked Cable and High-Force Tractor

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Abstract

Accessing ultradeep reservoirs for surveillance and intervention poses significant challenges when using conventional methods. Navigating complex trajectories with long downhole tool assemblies in ultradeep wells requires considerable conveyance forces and extreme wireline surface tensions that surpass the capacity of standard wireline logging units. This paper addresses the conveyance challenges encountered by an operator in the United Arab Emirates (UAE) and outlines an innovative combination of solutions that significantly enhance wireline accessibility to unprecedented depths. The approach involves overcoming conveyance challenges using a polymer-locked cable, dual-drum capstan, and high-force tractors. This ensures safe access and retrieval of the bottomhole assembly (BHA) without damaging the cable while acquiring reliable data in deep reservoirs. The methods employed include:

- Analyzing well tortuosity to estimate surface tensions and conveyance forces.
- Selecting a conveyance solution that combines polymer-locked cable, high-force tractor, and dual-drum capstan.
- Testing cable friction with the capstan.
- Training wireline crew on capstan setup and operation for production logging.
- Performing surface integration tests to evaluate pulling capacity, speed, and surface friction.
- Conducting risk assessments and implementing control measures.

The innovative integration of conveyance solutions enabled access to an unprecedented depth of 37,725 ft within an ultradeep horizontal well. Employing the capstan mitigated the risk of excessive cable tension at the cable drum, thereby preventing potential damage to both the cable and the drum. This conveyance methodology enabled the operator to fully access the well and obtain production logging data efficiently. This marks the first deployment of a polymer-locked cable using a dual-drum capstan and high-force tractor. It represents the world's deepest tractor conveyance using capstan technology. The success of this deployment demonstrates how innovative technology can address and overcome conveyance challenges

associated with conventional methods, thus enabling reliable access to ultradeep reservoirs. It allows operators to extend wireline intervention to reach long horizontal wells up to 40,000 ft measured depth (MD) and beyond.

Introduction

An oilfield operator in the UAE is redeveloping the second-largest offshore oilfield in the world, which holds an estimated 50 billion barrels of oil. The redevelopment incorporates advanced technologies, including the construction of four artificial islands that function as drilling bases, facilitating the drilling of deep wells and a central processing facility. This approach reduces drilling costs by enabling the use of land rigs instead of jack-up or semi-submersible rigs. These islands can support up to 2,000 slots for drilling. Drilling numerous wells from a limited area poses a challenge in avoiding intersection with previously drilled wells, resulting in more complex well trajectories with multiple bends and turns. To extract oil from reservoirs near the field's periphery, extremely long laterals must be drilled. Advancements in drilling technology have enabled the drilling of some of the world's longest and most complex wells in this field, including the world's longest well, which has a measured depth of over 52,000 ft. The horizontal laterals in these wells extend up to 25,000 ft and beyond.

A typical well in this field has an upper completion comprising 4.5-in. tubing with a minimum restriction of 3.8-in. The lower completion in these wells comprises 6 5/8-in. casing in an 8.5-in. open hole. Multiple swell packers set in the open hole along the lower completion casing divide the pay zone into multiple compartments. Fluid entry into these compartments is through downhole inflow control devices (ICDs) or pre-perforated liners (PPLs) with 1-in. holes, or a combination of both. The ICDs regulate and balance inflow from each compartment using mechanisms such as nozzles, tubes, helical channels, or chambers. Operators can adjust production by controlling the sliding sleeves on the ICDs. (Varghese et al. 2020) Fig. 1 depicts the typical completion schematic of an extended reach well in this field.

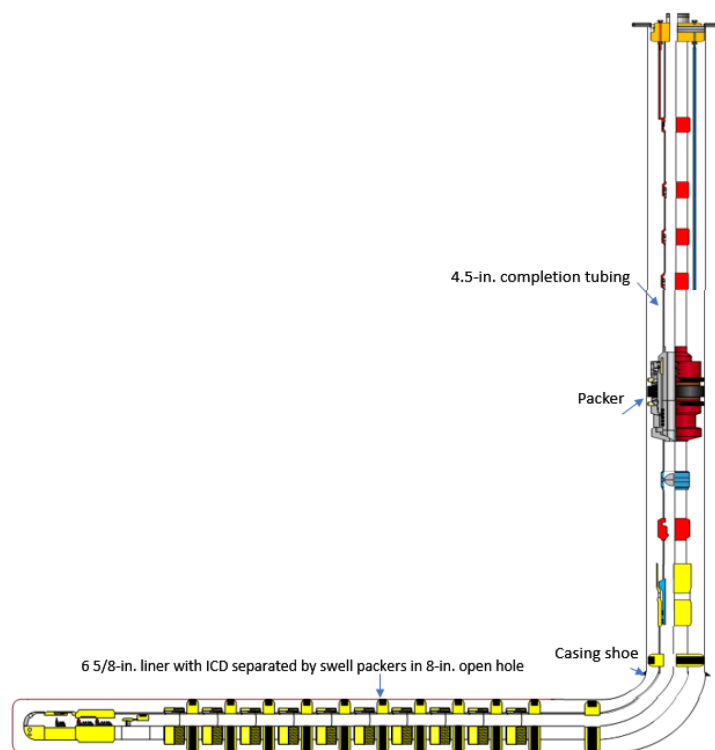


Figure 1—Schematic of a typical completion in an extended maximum reservoir contact well. (Varghese et al. 2020)

Drilling horizontal Extended Reach Drilling (ERD) wells serves to maximize reservoir contact and significantly enhance well productivity. While the cost per individual well increases, the overall drilling capital expenditure can be optimized by reducing the total number of wells required. Consequently, well surveillance is essential for assessing production performance, maintaining well-integrity, and monitoring for gas and water breakthrough. As this field notably contributes to the UAE's total oil production, comprehensive well surveillance is critical for effective production management and prevention of operational disruptions. Furthermore, well intervention is necessary to remediate water or gas breakthrough and to manipulate sliding sleeves of Inflow Control Devices (ICDs) to optimize production performance. Recent advancements in drilling technology have enabled the drilling of complex wells that are longer than ever before; however, employing conventional conveyance methods for surveillance and intervention in these wells remains challenging.

Well-surveillance involves using wireline cables to lower downhole logging tools equipped with sensors that measure parameters such as pressure, flow, temperature, saturation, fluid holdup, and tubing/casing diameters into the wellbore. The collected downhole data allows for quantifying fluid contributions, identifying water breakthroughs, monitoring gas-oil-water contacts, and assessing integrity of the well completion. The wireline logging cable serves both as a mechanical means to position these tools within the well and as an electrical connection for power and data transmission to the surface acquisition system. In vertical wells, the weight of the tools typically enables straightforward conveyance via logging cable. However, in highly deviated or horizontal wells, gravity may not provide sufficient force, making tool conveyance more complex and necessitating alternative deployment methods. Coiled Tubing (CT) is a commonly used technique that permits lowering of the tools into the wellbore using a continuous length of steel pipe that is flexible enough to be spooled and unspooled from a reel. Tools can operate in memory mode, requiring only a mechanical link, or can provide real-time data through a wireline cable that is pumped inside the coiled tubing.

An alternative conveyance method involves the use of wireline tractors, which are powered robotic devices equipped with either wheels or reciprocating mechanisms, operated via electrical power transmitted through the wireline cable. In both configurations, the necessary traction for movement is achieved through direct contact between the traction mechanism and the wellbore walls. Wheeled tractors, noted for their simplicity, represent the most widely used type within the industry. A standard tractor operation involves lowering a Bottom Hole Assembly (BHA) composed of tractor tools with multiple wheel sections, along with logging tools attached beneath the tractor assembly, into the well using a wireline cable. The tool assembly is conveyed to the depth limit at which gravitational forces are insufficient to advance the tools further. At this juncture, the tractor tools are activated to continue transporting the assembly into the wellbore. These tractor tools not only propel the logging tools through deviated or horizontal well sections but also facilitate the movement of the wireline cable deeper into the well.

Most oilfield operators prefer the use of wireline tractor conveyance method due to the various benefits of improved efficiency, reduced equipment footprint, reduced personnel, precise depth control, improved measurement data quality and reduced emissions.

Historically, the operator encountered significant difficulties in obtaining data from wells exceeding 25,000 feet, primarily due to excessive tensions on the wireline cable that often surpassed the cable safe working limits, which restricted access for well surveillance. The success rates of full well access were only 30% using conventional technology. Many operations resulted in partial data gathering. The introduction of advanced high-strength, low-friction cables made it possible to reach depths of up to 33,000 feet. However, drilling beyond this depth introduces additional challenges, such as extreme surface tension resulting from extended cable lengths, greater power delivery requirements, and substantial tractor forces needed to deploy tools. Addressing these complexities requires a combination of diverse conveyance solutions operating collaboratively, rather than relying on a single method.

Conveyance Challenges

High Cable Tension

During well surveillance or intervention operations, the wireline cable plays a crucial role in retrieving tools from the wellbore. To ensure safe retrieval of the BHA, the cable must possess sufficient strength to counteract the combined forces generated by the weight of the tools, the weight of the cable itself, and the drag exerted on both the tools and the cable. In ultra-deep, deviated wells with complex trajectory bends, cable drag becomes a significant contributor to surface tension due to frictional interactions within the wellbore. As illustrated in Fig. 2, such complex wells exhibit variations in both deviation and azimuth relative to depth, resulting in the trajectory bending across all three spatial planes. Consequently, the cable is subjected to cumulative bending effects, with each bend serving as a potential capstan point, further increasing the overall tension encountered during operations.

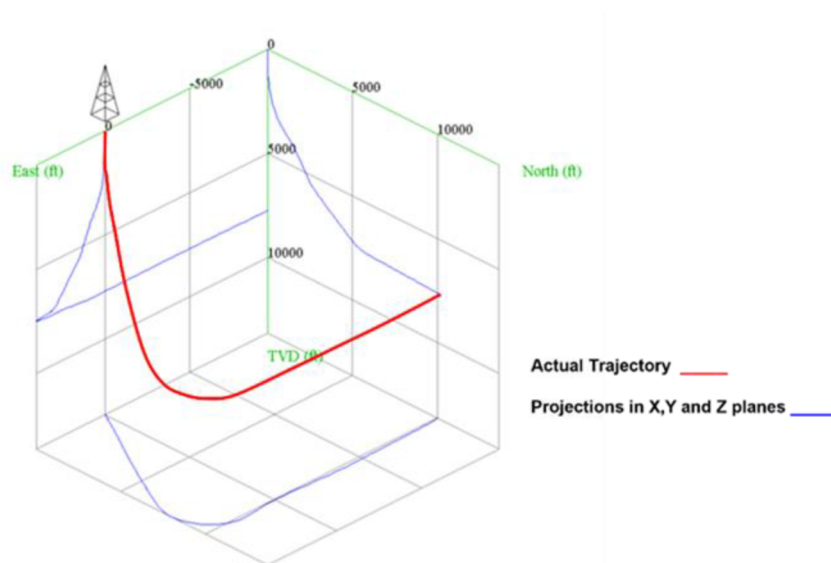


Figure 2—3D well trajectory of an example well with projection in X, Y, and Z planes.

When a cable wrapped around a bend is pulled, the cable tension interacts with the surface of the bend such that the force required to pull the cable is higher than the load. This is known as the capstan effect as shown in Fig. 3. The Euler-Eytelwein capstan equation shows the relationship between pulling tension, the holding tension, the friction coefficient, and the wrap angle.

$$T_{\text{Load}} = T_{\text{Hold}} e^{\mu\phi}, \quad (1)$$

where T_{Load} is the cable tension above the capstan point; T_{Hold} is the holding tension below the capstan point; μ is the friction between the cable and the wellbore; and ϕ is the wrap angle. (Varghese et al, 2024)

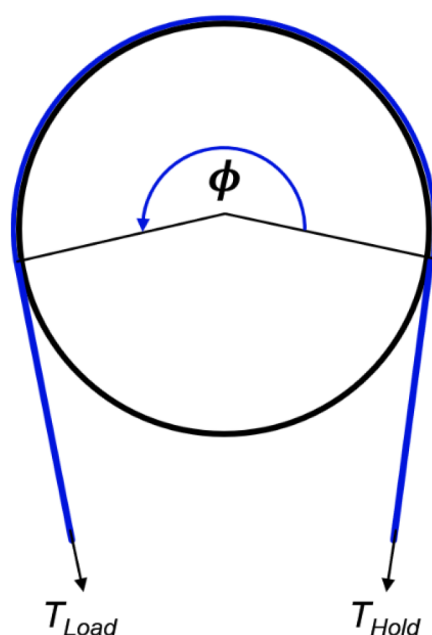


Figure 3—Capstan effect.

The tension along the cable increases significantly at each bend, resulting in extremely high surface tensions. To address this, high-strength cables with adequate safe working load must be utilized. However, enhancing cable strength typically increases both its weight and diameter. A larger cable diameter necessitates a greater tool weight to counteract the forces created by well pressure on the cable at the grease injection head during tool deployment, which can extend the BHA length and potentially exceed the available rig-up height for rigless operations. While additional cable weight may have minimal impact in shallow wells, it can lead to substantial surface tension increases in deep deviated wells as the holding tension value increases. Therefore, careful optimization of these two cable parameters is essential. An effective approach is to reduce cable friction by carefully selecting appropriate materials for the cable surfaces that interact with the wellbore. Since the relation between friction and tension is exponential, a slight change in friction can result in substantial reduction in surface tension. However, it must be noted that the static friction of cable and tools in the wells will be greater than the dynamic friction and thus the cable must have high strength to retrieve the tools safely assuming the static friction of the tools and cable. The ideal cable would be one with low friction, small diameter, and high strength.

Excessive Cable Drum Forces

Extreme surface tensions on cables in ultradeep wells are unavoidable. While cables with high tensile strength can meet the necessary load requirements, spooling these cables onto the drum presents additional challenges. When multiple layers of cable are wound onto the drum under significant tension, they exert compressive radial forces on both the cable and the drum itself. These issues are amplified with longer cables, as more layers add to the overall force exerted on the drum. The core of the drum experiences increased internal pressure, while the flanges are subjected to bending moments. Such conditions may lead to structural failure of the drum core, potentially causing it to collapse during operations. Bending moments can also generate cracks near the junction of the drum core and its flanges. Both failure modes present serious risks, including cable damage and an inability to retrieve the cable from the well, resulting in significant non-productive time. When a cable is wound onto itself, it must be able to withstand the cumulative transverse stresses created by multiple layers of cable spooled over it. The cable is designed to withstand a specific amount of pressure before deformation may occur. Excessive transverse stresses beyond this limit can result in cold flow, where the polymer inside the cable deforms, potentially allowing the metal armors to penetrate

the insulation and compromise the cable's electrical integrity. Typically, when the cable is wound on a drum, successive layers are placed between the grooves of the previous layer, providing mechanical support that helps prevent deformation. However, because the cable is spooled from end to end, continuous spooling in this manner is not possible; the cable must cross over the previous layer at certain points to change direction. This crossover portion is vulnerable to crushing due to fewer contact points. Managing these risks is an important consideration for wireline operations in ultra-deep wells. Hence it is essential to use a tension relief system that allows the cable to be spooled at reduced tension at the drum. Fig. 4 shows the various forces generated by spooling cable under tension on the cable drum.

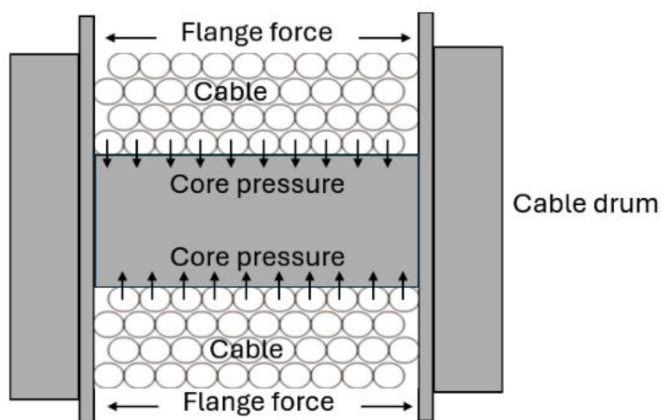


Figure 4—Forces acting on the cable drum caused by cable tension.

Fig. 5 shows an example of the deformation of the cable due to cold flow at the crossover zone. Note the deformation changes the circular shape of the cable. This can lead to spooling difficulties, difficulty in cable passing through the pressure control equipment and further cable damage.



Figure 5—Cable deformation on cable drum due to excessive spooling tension.

In summary, managing high tension in wireline cable winches requires careful engineering to prevent damage, maintain safety, and ensure efficient cable handling.

Extreme Tractoring Forces

Deploying extended lengths of cable into an ultra-deep well necessitates substantial tractoring force. The wireline tractor is responsible for conveying both the logging tools and the cable itself into the wellbore. As both components are subjected to frictional drag, the tractor must effectively overcome the combined resistance to ensure successful conveyance of the tools into the well. Typical wireline tractors equipped with multiple drive sections can generate a conveyance force of approximately 2100 lbf. In wells exceeding 33,000 ft in depth, the required conveyance forces range from 2500 to 3500 lbf. The required force reaches its peak when the tractor halts and restarts following an up-tension check in the deep deviated section of the well. In this scenario, it must overcome the higher static friction of the cable and tools as compared to dynamic friction. Therefore, the selected tractor should have adequate force capacity under downhole conditions to enable the transition of tools and the cable from static to dynamic states. Fig.6 shows the various forces required to convey the tools into the wellbore. The traction force produced by the tractor must exceed the combined forces of cable drag, tool drag, and fluid lift.

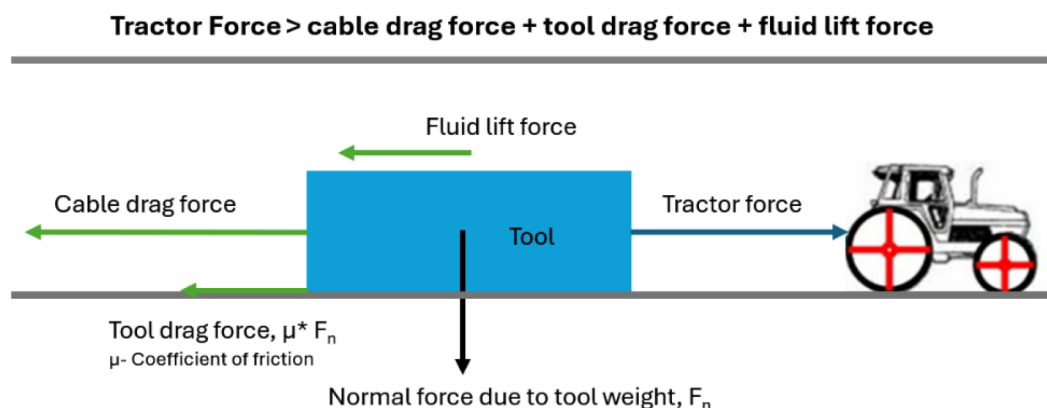


Figure 6—Tractor force required to convey tools. (Varghese et al. 2024)

Cable power delivery limitation and telemetry

Ultradeep wells require extremely long cable lengths. The cumulative resistance of a long length cable can limit the amount of power that can be delivered downhole. Increased tractoring forces result in a need for more power to be transmitted through the cable. We are presented with the challenge of delivering increased power over an extended cable length. The longer cable also exhibits greater capacitance, resulting from the cumulative effect of its length, which in turn limits the available data rate for the cable telemetry system. Maintaining a sufficient data transmission rate is essential both for effective tractor tool control and for acquiring high-resolution data from production logging tool sensors.

In summary, ultradeep wells with complex trajectories present various well conveyance challenges for well access, including extreme tensions, high cable drum forces, significant tractor force needed to convey tools into the wellbore and limited power delivery and telemetry over long cables. Addressing these challenges requires the application of multiple technical solutions.

Cable Tension Analysis

The operator required to acquire production logging data from a 38000 ft long horizontal well. Previously this well was accessed up to 30000 ft due to the limited length of cable available. The oil production from compartments beyond this depth could not be acquired due to the partial access. This time production logging data beyond 33,000 ft was required. A team of well operations specialists and conveyance experts

was formed to develop solutions for wells over 33,000 ft MD. They assessed survey, completion, and production data for the wells, then used simulation software to model conveyance and tractor forces. Simulations indicated that the estimated tracting forces were approximately 2,500 lbf. Both dynamic and static cable friction were accounted for in calculating the cable tensions. Given the high sensitivity of cable tension to friction, it was imperative to utilize cables with minimized frictional properties. Nevertheless, even with low-friction cables, some uncertainty remains due to variables associated with completion landing, which can affect accessibility. Managing completions in extremely deep wells with extended lengths presents challenges, particularly in keeping the completion aligned with the wellbore. There are no mechanisms available to pull the completion into position while lowering it, and multiple bends along the well trajectory may result in sinusoidal or helical buckling. Such buckling can be exacerbated by thermal expansion caused by transitions from flowing to shut-in conditions, leading to an increase in completion length and, consequently, additional doglegs along the cable path. It should also be noted that deviation surveys are typically recorded in open hole conditions, which may differ from those within the final completed well; thus, actual in-well cable tensions could deviate from predicted values. Accordingly, it is essential that the cable possesses sufficient Safe Working Load (SWL) margin to facilitate tool retrieval in scenarios involving elevated cable drag. Additionally, the tracting forces necessary for conveyance may also be influenced by these factors. Therefore, it is essential to incorporate an adequate margin for tracting force requirements. Fig. 7 shows an example of a tractor force simulation using static friction coefficient and dynamic friction coefficient.

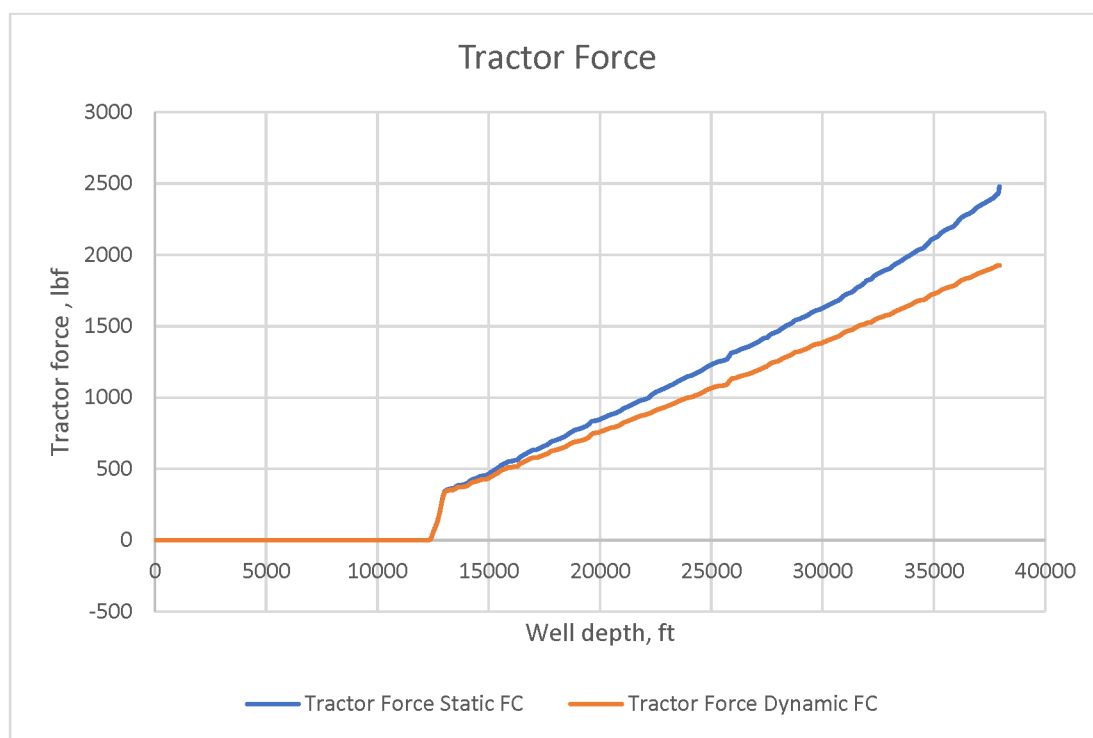


Figure 7—Example of tractor force and cable tension simulation results.

Conveyance Solutions

Cable

A polymer-locked wireline cable with an outside diameter of 0.44-in. and a length of 45,000 ft was selected for this operation. This cable has previously been deployed successfully in multiple wells to a maximum depth of 33,000 ft. The polymer-locked cable features a smooth jacket, with its outer surface fully covered

by a polymer layer. The armor wires are embedded within the polymer, ensuring that the cable's contact with the wellbore is through the polymer rather than exposed metal armors, as is typical with conventional wireline cables. Furthermore, this cable features a torque-balanced armor package and exhibits resistance to rotational forces under load. The torque-balanced design enhances the cable's strength-to-weight ratio, enabling the construction of high-strength cables with optimized outer diameters. For this operation, a cable with a safe working load (SWL) of 11,000 lbf. has been selected. This cable is engineered with a gas-resistant construction, utilizing advanced polymer processing and bonding methodologies in conjunction with a composite, smooth outer polymer jacket. The manufacturing process ensures that the polymer completely occupies the spaces between both the central copper strands and the surrounding copper strands, thereby creating an effective gas-blocking core. Additionally, the polymer fills the void spaces in the armor layers and secures the armor wires within the polymer matrix. The armor matrix's layered design plays a crucial role in achieving torque balance within the cable. The presence of polymer in the armor matrix increases friction between the armor wires, effectively limiting their movement when the cable bends over sheave wheels. This characteristic assists in preserving the cable's safe working load (SWL) during such operations. This configuration not only supports torque balance but also prevents the armor wires from becoming stranded. As a result, cable twisting and seasoning are unnecessary, making this design especially advantageous for ERD wells. Fig. 8 illustrates the cross section of the 0.44-in. polymer-locked cable. This cable features a 13AWG copper conductor ensuring higher electrical power delivery required for tractor operation.

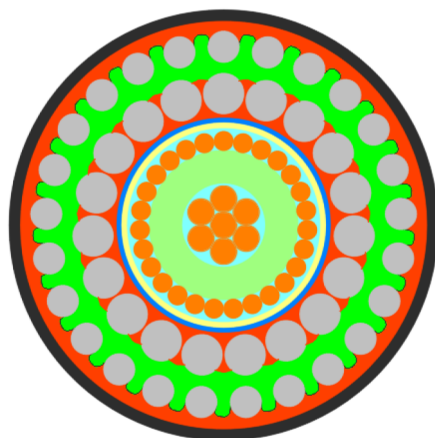


Figure 8—Cross-section of smooth-jacketed, polymer-locked cable.

Operational data indicate that the composite polymer jacket reduces drag force by up to 40% compared to conventional cables. This allows for increased well-depth access, faster deployment, and higher overpull capacity at the surface (Varghese et al. 2020). As a result, this cable has the potential to substantially improve the efficiency of logging operations.

Furthermore, the cable's smooth exterior allows compatibility with a rubber packoff for well pressure control, providing an alternative to conventional grease injection heads used with standard cables. This facilitates greaseless operation, resulting in a cleaner wellsite, eliminating the risk of grease spills, and reducing rig-up height. Additionally, reduced friction enables higher cable speeds during operation.

Wireline Dual-drum Capstan

Spooling long cables under high tension exerts significant pressure on the cable drum core and induces bending moments at the drum flanges, potentially resulting in damage to the drum. Even when utilizing a reinforced cable drum, the compressive forces generated can lead to cable deformation and cold flow. Surface tension-relief systems are designed to mitigate these challenges: they deliver primary pulling capability exceeding that of the winch unit and significantly reduce cable tension during storage on the

drum. These measures extend the service life of both the cable and drum, prevent core crushing due to cold flow, and enhance safety for personnel in the vicinity.

The common options for tension relief are powered sheaves and wireline dual-drum capstan winch. The powered sheaves have limited pulling capacity and are not preferred. The wireline dual-drum capstan has several advantages over the powered sheave like higher pull, better distribution of load over the sheave. The dual-drum capstan winch consists of two hydraulically powered sheave drums with multiple grooves. The two drums are operated with separate hydraulic motors. The two drums are aligned in such a way that the cable exiting from one drum has a straight path into the groove of the next drum. The dual-drum capstan tension relief mechanism operates on the principle of the Euler-Eytelwein equation, also known as the Capstan equation, as shown in Fig. 9.

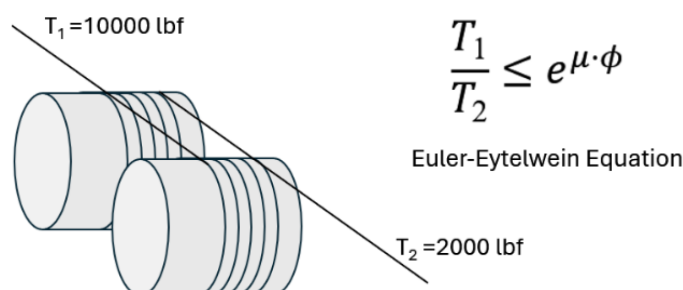


Figure 9—Capstan winch principle. (Varghese et al.2024)

Whenever a cable under tension is wrapped on a drum, the cable tension and friction interact to create a force reduction from one end to the other. This force reduction depends on the angle of wrap and the friction between the cable and the sheave drum. By increasing the number of turns, sufficient reduction in tension can be achieved. The wireline dual-drum capstan is deployed at the surface between the logging unit winch drum and the upper sheave. It can either be used in conjunction with a lower sheave or replace the lower sheave while providing the principal pulling capability. The differential reduction in tension across the capstan enables safe spooling and storage of the cable on the winch cable drum at a reduced tension. The function of the wireline dual-drum capstan is similar to the Coiled Tubing Injector head which enables safe spooling of the coiled tubing on the reel at reduced tension. Fig. 10 shows a typical wireline dual-drum capstan used in wireline operations.



Figure 10—Wireline dual-drum capstan unit. (Varghese et al.2024)

The wireline dual-drum capstan is mainly used for wireline logging in deep or challenging wells. Its application in cased hole operations, particularly during rigless operations with pressure control equipment in producing wells, is rare. This is due to several factors. High-tension operations are classified as high-risk, and the use of a capstan may introduce various HSE risks, including the potential for cable damage if appropriate planning and precautions are not implemented. Cable damage in high-pressure, high-tension situations can disrupt operations and may leave a broken cable in the well, impacting production and future intervention if not retrieved. Fishing the cable is time-consuming, costly, and may cause extensive downtime. Also, previous generation cables used for cased-hole operations were limited in strength and did not require the use of tension relief systems. However, with the advent of high tension coaxial and monocables, this need becomes imperative. Hence for this operation, a wireline dual-drum capstan was selected. The wireline logging unit was modified for wireline capstan operation. The primary winch control from the unit is routed to the capstan winch while the cable drum is maintained at a constant set tension automatically through electronic control. Separate tension devices measure the cable tension at the low-tension drum side and the high-tension capstan side. These tension devices are interfaced with a control unit to provide feedback for automatic tension control. Fig.11 shows the schematic of the various interfaces between the components used for wireline capstan set up with the wireline logging unit.

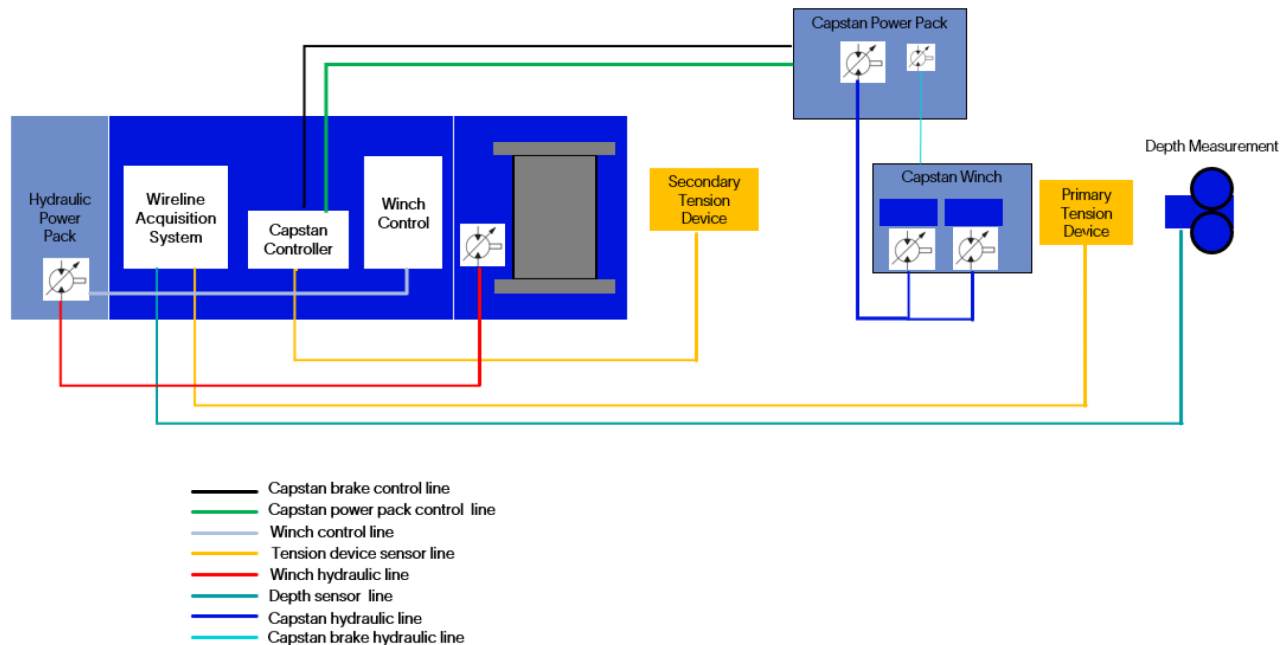


Figure 11—Capstan control interface with logging unit

Experience with dual-drum capstans in cased hole environments is currently limited; consequently, thorough planning and multiple surface integration tests were essential. A key challenge was reduced friction between the lubricated polymer-locked cable and the dual-drum capstan. As previously noted, force reduction depends on friction and the number of cable-turns, with typical dual-drum capstans allowing a maximum of five turns. Typically, a friction coefficient of 0.125 is applied to hepta-cables used in open hole logging operations to calculate the number of turns on the capstan. In this instance, however, the cable is equipped with a low-friction polymer jacket rather than conventional metal armoring. It should be noted that there is no prior field experience involving the use of a polymer-jacketed cable with a capstan. It was therefore crucial to determine the maximum tension reduction achievable under these lower-friction conditions. Exceeding the allowable differential tension across the capstan could cause significant cable damage or even lead to cable failure. To address this issue, a test was conducted utilizing a polymer-locked

cable with a Safe Working Load (SWL) of 11,000 lb. The purpose of this assessment was to verify the minimum number of cable wraps required on the capstan grooves to prevent slippage, while ensuring that a differential tension of at least 9,000 lbf across the capstan was maintained. The test setup incorporated two spooling units to transfer cable from one to the other. As illustrated in Fig. 12, two wireline dual-drum capstan units were employed: one maintained high tension, while the other was utilized to evaluate slippage. The objectives of the test were as follows:

- To determine through experimental testing the coefficient of friction behavior between a lubricated polymer locked cable and capstan's metal grooves when system is subjected to defined number of wraps and differential cable tension.

- Calculate dynamic coefficient of friction using the relation: $\mu = \frac{\ln\left(\frac{T_1}{T_2}\right)}{\phi}$

μ = Coefficient of friction

T_1 = High Tension

T_2 = Low Tension

ϕ = Angle made by wraps on capstan in radians (1 wrap = 2π rad)

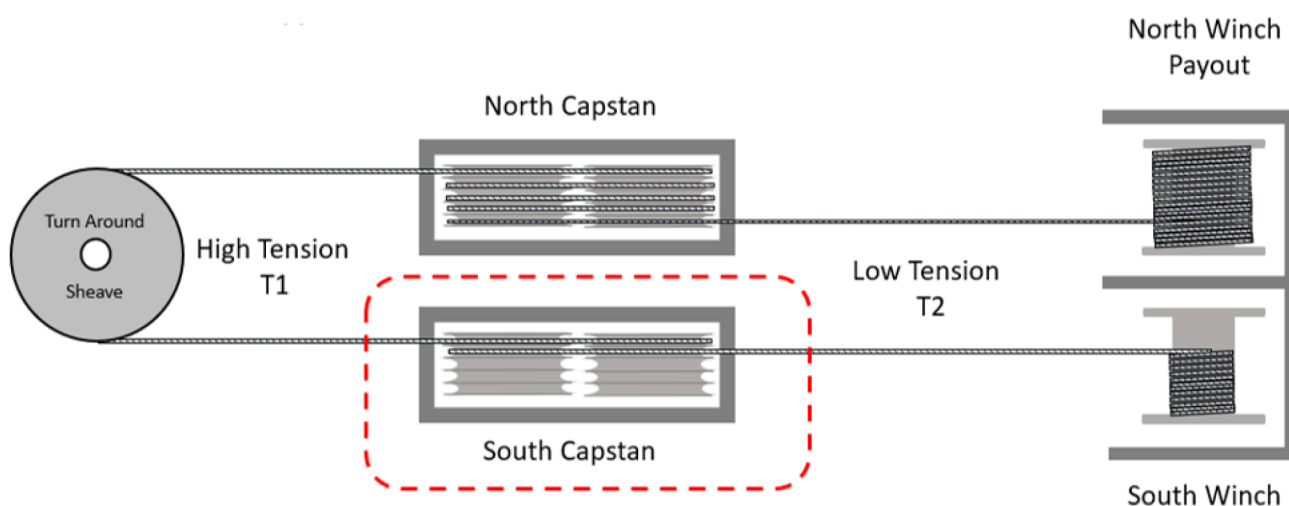


Figure 12—Test set up for assessing slippage with polymer locked cable over capstan.

Tests were conducted by gradually reducing the number of cable wraps to identify the minimum turns necessary to achieve a differential tension of 9,000 lbf, beginning with 2,000 lbf at the cable drum and reaching 11,000 lbf on the high-tension side. The test results indicated that at least four turns were needed to prevent slippage, as cable slippage occurred with only three turns. Consequently, all five available turns on the capstan were used for this operation. Fig. 13 shows the comparison of the test results using 3 and 4 wraps, respectively.

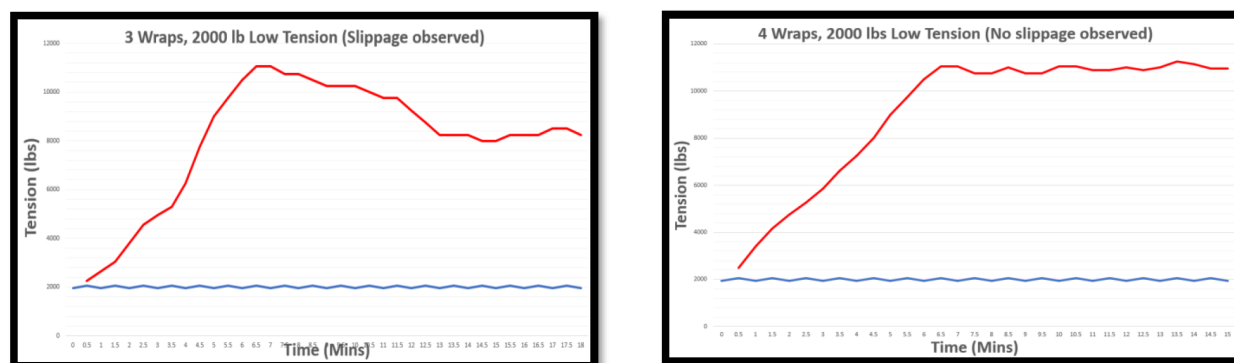


Figure 13—Slippage test results with 3 and 4 wraps

High-Force Wireline Tractor

To address the requirements associated with higher-than-normal tractor forces, the team selected a 3-3/8" OD, high-force tractor capable of providing a cumulative force of 3600 lbf using six drive sections with two wheels each. While this high-force tractor had been used for shallower operations previously, this deployment marked its first use at depths beyond 30,000 ft. Fig. 14 presents the high-force tractor tool. The high-force tractor is designed to enable faster operation, maintain consistent speeds, and allow data acquisition while tracting tools into the well. The high-force tractor incorporates an advanced generation telemetry system capable of uplink data rates reaching up to 700 kbps. This level of data bandwidth is distinctive within the industry for coaxial and monocable data telemetry systems. It enables high-resolution data acquisition during tractor operations from both tractor tools and the production logging sensors located below the tractor. Capable of delivering up to 600 lbf/drive in traction force, this bidirectional tractor also offers a contingency for releasing stuck tools through reverse tracting, with a total maximum downhole pull of 3600 lbf in addition to the cable pull. This function addresses situations in ultradeep wells where available pull at significant depths may be insufficient to release stuck tools.



Figure 14—High-force tractor tool.

Operational challenges and solutions

There were several operational challenges to integrate all these technologies together. While these technologies were tested individually for deployment, this was the first time when all these innovations were incorporated into the same operation. Also, this was the first deployment for the wireline capstan winch in this environment. The first challenge was to accommodate 45,000 ft of cable drum on the existing logging unit. Spooling long cables require larger drums which require special logging units which can accommodate larger drums. These logging units have longer lead times for procurement hence this was a challenge for this operation. The team decided to incorporate design changes into the existing logging winch in consultation with the logging unit manufacturing center. New parts required for the modification were procured and manufactured locally improving on the lead times. The large cable drum was successfully installed onto the modified winch unit and subsequently tested to verify its performance under the specified load requirements. The modified winch still maintained the DNV certification of the structure thus allowing the transportation of the spooled cable drum with the winch. This eliminated the need to install the cable drum at the wellsite

and decreased the health, safety, and environmental risk associated with this activity, thereby enhancing operational efficiency.

The subsequent task involved anchoring the wireline dual-drum capstan unit. In contrast to offshore operations, where the capstan unit is attached to the rig or platform structure by welding or bolting, this land-based operation lacked a fixed structure for anchoring. As a result, the team designed and manufactured a platform frame weighted with dead weights to create sufficient friction with the ground for anchoring the capstan unit. The capstan was then bolted onto the frame for a secure attachment. The assembled platform frame, including the capstan and dead weights, was tested during a surface integration test with a pull of 11,000 lbf to confirm stability during operation. Following this test, the total weight was increased to 23 tons to provide an adequate safety margin in case ground friction decreased due to wet conditions. Fig. 15 shows the capstan installed on the platform frame.



Figure 15—Capstan winch installed with a tilt on the platform frame with dead weights.

Normal open hole operations typically employ a heavy BHA, resulting in consistently high tension when lowering tools into the well. In contrast, this specific operation in a horizontal well involved significantly lower surface tensions ranging from 500 to 700 lbf during tractor activities. Operating with such reduced surface tension using the capstan is uncommon and can present cable slippage challenges. Consequently, a System Integration Test (SIT) was performed to replicate running in hole, tractor activities, and pulling out of hole conditions with a surface setup comprising two spooling units. This SIT allowed for the determination of appropriate drum tension, speed, and other winch parameters required for the operation.

The SIT also identified the possibility of cable slipping from the capstan, which could occur due to insufficient tension at the cable drum side, changes in the cable entry position as the drum transitions from fully wound to partially wound, and the lateral movement of the cable on the drum during spooling. This slippage could potentially damage the jacket or the armors of the cable. To address this, the frame design was adjusted to tilt the capstan by 5 degrees, allowing the cable to enter directly into the high-tension sheave of the capstan. Fig. 16 depicts the slippage risk at the low-tension side. Fig. 17 shows the tilted frame design that tilts the capstan by an angle of 5 degrees to allow the cable to enter directly into the high-tension sheave. Fig. 18 shows the proper entry of cable into the capstan.

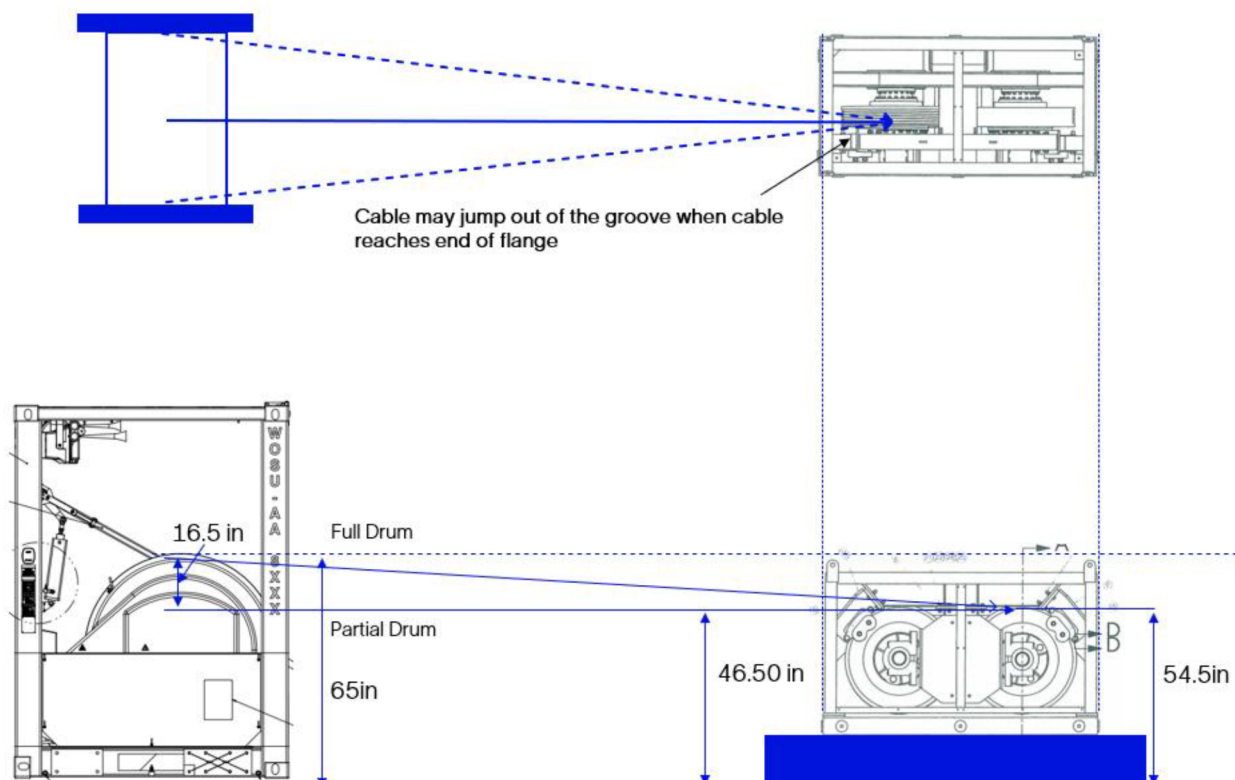


Figure 16—Risk of cable slippage resulting from a change in cable position.

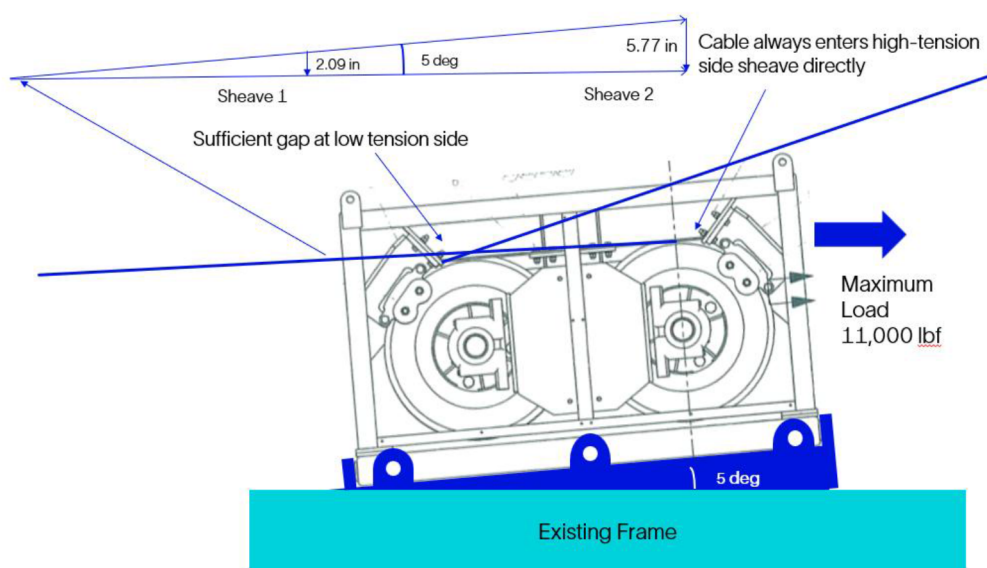


Figure 17—Tilted frame design that ensures cable does not touch the low-tension side sheave at entry.



Figure 18—Capstan alignment with proper entry of cable into the high-tension side groove

Capstan operations present significant risks, and given their rare occurrence in the Middle East, it was imperative that the field crew developed sufficient expertise to execute these tasks safely. Accordingly, engineers and operators received hands-on training from certified trainers and subject matter experts. This training included simulating the different scenarios of running the cable in the well, tractor operation and pulling the cable out of the well. Given the distinctive nature of this operation, additional risk mitigation measures were implemented, such as limiting the cable winch pull by adjusting the hydraulic pressure relief valve on the logging winch. This precaution helps prevent single-point failures that could result in unintended cable winch pulls.

Proper alignment of the cable with the sheaves and capstan is essential to ensure safe deployment. Accordingly, the crew conducted a thorough survey of the wellsite prior to operations to evaluate space availability for rig-up activities and precise equipment positioning. During deployment, a laser device was utilized to verify accurate entry and exit angles between the capstan and cable sheave wheels. Fig. 19 shows the alignment of the unit, dual drum capstan unit and the lower sheave in a straight line. The field team, in collaboration with subject matter experts and the operator well engineering team, performed a comprehensive risk assessment and implemented necessary control measures to mitigate or eliminate identified hazards.



Figure 19—Logging unit, dual drum capstan and lower sheave alignment at wellsite

Presentation of Data and Results

The innovative integration of conveyance solutions comprising the 0.44" OD polymer locked cable, high-force wireline tractor and the wireline dual-drum capstan winch enabled access to an unprecedented depth of 37,725 ft within this ultradeep horizontal well. Production logging tools, comprising an array of flowmeters, gas, and water holdup probes, two axis calipers as well as pressure, temperature, and well-deviation sensors, were efficiently deployed to the target depth within the well. The total BHA length that was used including the tractor was 93ft. The application of the 0.44" OD polymer-locked cable effectively minimized friction and delivered substantial pulling force, facilitating the safe retrieval of the tools with a maximum pull recorded at 9,200 lbf. The high-force tractor successfully conveyed the tools from 4,000 ft to 37,725 ft, resulting in a total tractoring distance of 33,725 ft, at an average operational speed of 1,200 ft/hr. This resulted in considerable time savings compared to previous interventions. Utilization of the capstan significantly reduced the risk of excessive tension on the cable drum, mitigating potential damage to both the cable and drum assembly. The cable was spooled at the drum at a tension range of 2500 to 3000 lbf throughout the operation. This conveyance methodology enabled the operator to access the well completely and obtain production logging data in an efficient manner. Data acquisition during tractoring operations yielded high-quality measurements from both the flowmeter and hold-up probe, while also eliminating the need for multiple passes along the extended horizontal section. Continuous monitoring of caliper readings beneath the tractors mitigated the risk of tool sticking due to potential reductions in completion internal diameter caused by scale buildup or damage. Given this achievement, it is evident that the methodology can be effectively applied to access wells exceeding depths of 37,725 feet.

Key achievements during this operation include:

- Accessing reservoirs at a depth of 37,725 ft, the deepest conveyance using a dual-drum capstan and tractor
- Achieving a cumulative tractoring distance of more than 33,725 ft in a single pass
- Acquiring downhole production logging data reliably over a 43,500-ft-long polymer-locked cable
- Simultaneous acquisition of production log data during tractoring, ensuring data quality and operational efficiency
- Quantifying horizontal lateral zonal contributions from all compartments of the lower completion.
- Surface and downhole tension data analysis to determine the friction reduction offered by polymer-locked cable and predict maximum reach possible with low-friction cable, high-power tractor, and capstan system for future operations.

Fig. 20 shows the comparison between the predicted tensions from the simulations versus the actual tension in the well. The measured dynamic tensions align closely with the predicted values at a cable friction coefficient of 0.2, illustrating the polymer-locked cable's effectiveness in reducing friction. It is noteworthy that significant tension of 9200 lbf was recorded at the greatest depth when transitioning the cable and tools from static to dynamic conditions. This demonstrates that, when planning such operations, it is essential to account for static friction in addition to dynamic friction, as this influences maximum surface tension and necessitates a higher Safe Working Load (SWL) for the cables.

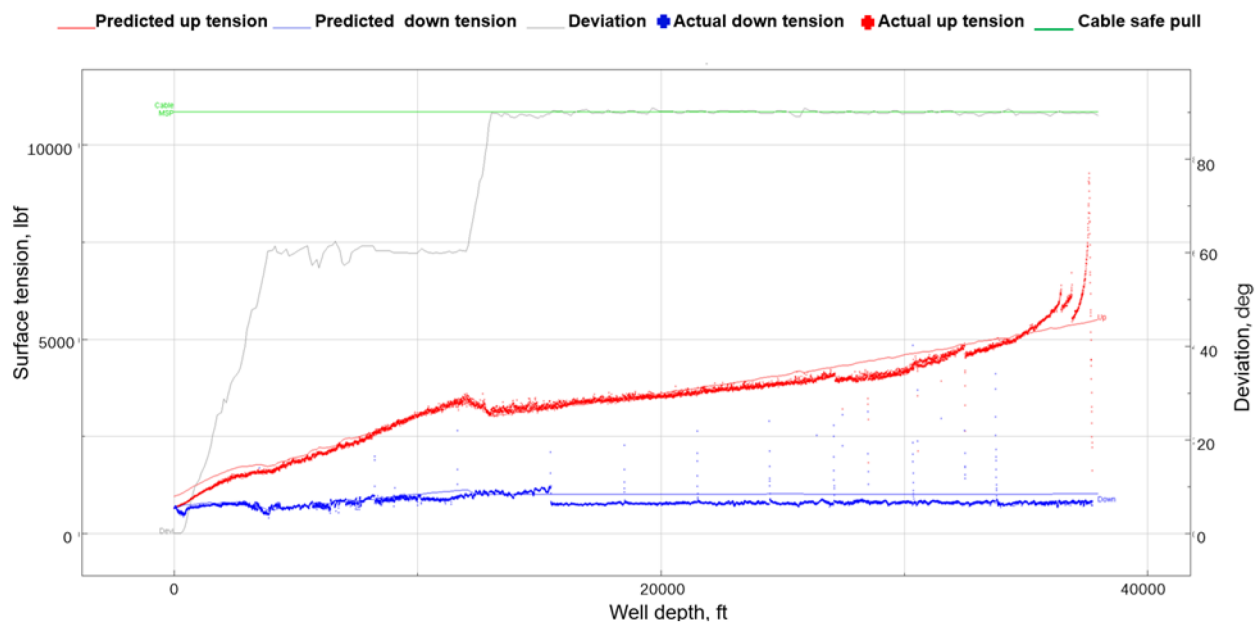


Figure 20—Actual surface tension comparison with predicted tensions

This operation marked the first deployment of a wireline capstan winch in a land-based rigless operation with pressure control equipment. The use of the platform frame with dead weights ensured that the capstan equipment did not move or slip on the ground and provided a stable anchor even at high cable tensions. Thorough job planning and comprehensive risk mitigation strategies were essential to this successful deployment, demonstrating that these challenges can be effectively managed to ensure the safe execution of high-risk operations. There were no safety or service quality incidents during this operation. This successful deployment shows that deep reservoirs can be reliably accessed for data acquisition and intervention, enabling efficient operations.

Conclusions

In summary, the successful integration of the polymer-locked wireline cable, high-force tractor, and wireline dual-drum capstan has redefined the boundaries of well access in ultradeep and extended reach drilling environments. This pioneering approach not only enabled the operator to overcome formidable challenges such as extreme cable tension, complex well trajectories, and power delivery limitation, but also set a new benchmark for operational efficiency and safety. By achieving reliable data acquisition at record-breaking depths and ensuring the integrity of both equipment and well assets, this methodology paves the way for more frequent, efficient, and sustainable well surveillance and intervention. Ultimately, these advancements signal a transformative step forward for the oil and gas industry, opening new possibilities for maximizing reservoir performance and extending the productive life of complex wells.

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